NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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13. ABSTRACT

Describes a method for evaluation of full-tracked carrier physical and operational performance characteristics. Identifies supporting tests, facilities, and equipment required. Provides procedures for test procedures, initial inspection, servicing, vehicle characteristics, safety, endurance, durability, and reliability.

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14. KEY WORDS	LINK A		LINK B		LINK C	
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Tracked Vehicles*						
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Personnel Carriers*						
Cargo Carrier (Vehicle)						
Armored Personnel Carriers*						
Vehicle, Tracked						
*New						

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AMSTE-RP-702-101
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CARRIERS, FULL-TRACKED (AUTOMOTIVE)

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SECTION I
GENERAL

1. Purpose and Scope.

a. This TOP provides guidance for planning tests of full-tracked carriers (automotive) to assure their conformance with Required Operational Capabilities (ROC's), MN's, TC's, and other guidance documents. Appropriate subtests suitable to meet the requirements of development tests I, II, or III (formerly developmental suitability tests, engineering tests, and initial production tests) can be selected from those listed in section II.

b. The compatibility of the carrier with any special weapon, such as a mortar, for which it is to be a platform is not covered in this TOP.

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2. Background. A carrier, full-tracked is a self-propelled, full-tracked vehicle designed primarily to transport personnel, cargo, or weapon systems over marginal terrain. Military carriers are designed to fulfill numerous needs and are classified by function, including:

a. Carrier, amphibious - a tracked vehicle designed to operate and carry a full cargo both on land and while floating and swimming on water.

b. Carrier, command post - a tracked vehicle designed to provide the unit commander with a mobile command post. The vehicle is used with other mechanized equipment such as carrier, armored personnel.

c. Carrier, armored personnel, full-tracked - a tracked vehicle designed to transport infantry troops to combat and provide protection from enemy small arms.

d. Carrier, cargo, full-tracked - a tracked vehicle designed to transport cargo. It usually has a flat loading platform, tailgate, and tarpaulin.

In addition to the above listed vehicles the carrier, full-tracked is used as a weapon mounting platform for various missiles, mortars, and flame devices.

3. Equipment and Facilities. Equipment and facilities are covered in the references of section II.

SECTION II
TEST PROCEDURES

4. Supporting Tests.

a. Subtests to be considered in formulating a development test plan, with TOP/MTP and other references, are listed below. The list is generally in a preferred order of completion; factors to consider are discussed in paragraph 5c.

TEST SUBJECT TITLE	PUBLICATION NO.
(1) Preparation for Test (refer to para 5)	
(2) Automotive.	
Initial Inspection (refer to para 6)	2-2-502
Stowage	2-2-802
Vehicle Characteristics (refer to para 7)	2-2-500
Preliminary Operation	2-2-505
Safety Evaluation (refer to para 8)	2-2-508
Braking, Tracked Vehicles	2-2-627
Steering	2-2-609
Center of Gravity	2-2-800

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TEST SUBJECT TITLE	PUBLICATION NO.
Gradeability and Side Slope Performance	2-2-610
Toxic Hazards	2-2-614
Winches	2-2-712
Acceleration - Maximum and Minimum Speeds	2-2-602
Standard Obstacles	2-2-611
Night Performance of Combat Vehicles	2-2-616
Electromagnetic Interference (Noncommunications Equipment)	2-2-613
Electromagnetic Emissions (Electronic Equipment)	*6-2-542
High and Low Temperatures	*2-2-816
Engine and Power Train Cooling Systems	2-2-607
Cold Starting and Warmup	2-2-650
Rain and Freezing Rain	2-2-815
High Altitude Effects	2-2-702
Load Distribution and Ground Pressure	2-2-801
Vehicle Fuel Consumption	2-2-603
Fuels and Lubricants	2-2-701
Tracks	2-2-705
Tracked Vehicle Suspension Systems	2-2-714
Towing Resistance	2-2-605
Drawbar Pull	2-2-604
Fording	2-2-612
Noise	3-2-811
Soft Soil Vehicle Mobility (Prepared Sites)	2-2-619
Off-Road Mobility (Natural Sites)	As determined by ROC, MN, test directive, and available terrain.
Field Shock and Vibration	2-2-808
Security from Detection (Vehicles)	2-2-615
Infrared Emissions from Vehicles	2-2-812
Communication Equipment (Vehicle)	2-2-709
Air Conditioners	2-2-713
Vehicle Heaters - Personnel	2-2-708
Electrical Systems (Automotive)	2-2-601
Solar Radiation	4-2-826
High Humidity	4-2-820
Fungus Resistance	4-2-820
Salt Water and Salt-Laden Air	2-2-520
Logistics-Over-The-Shore (LOTS)	2-2-520
Transportability	1-2-500
Cargo Loading Adaptability	2-2-537
Field Tests of Automotive Engines	2-2-721
Radioactive Components	3-2-711
Air Transportability	2-2-512 (until superseded by 1-2-500

*When published.

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TEST SUBJECT TITLE	PUBLICATION NO.
Vehicle Collision and Accident	2-2-621
Endurance and Reliability (refer to para 9)	2-2-507, 2-1-001
Durability (refer to para 9)	1-2-502, 2-1-001
Human Factors Engineering	2-2-803
Maintenance Evaluation	1-2-501, TECR 750-15
Value Engineering	In accordance with current directives
Overload Testing (Vehicle)	2-2-626

(3) Armament.

Stabilization Systems	3-2-602
Gun Control Systems	3-2-603
Boresight Retention	3-2-604
Accuracy Firing of Vehicular Mounted Weapons	3-2-605
Superelevation-Range Relationship of Ballistic Mechanisms	3-2-700
Night Vision Devices	3-2-706
Inspection and Measurement of Cannon	3-2-800
Measurements of Cannon	3-2-802
Noise and Blast Measurements	3-2-811
Optical Collimation of Range Finders	3-2-814
Field of Vision - Vehicles	3-2-812
Field of Fire	3-2-813
Grenade Launchers	3-2-030
Security from Detection (Weapons)	*3-2-520
Recoil and Equilibrator Systems	3-2-600
Fire Control for Air Defense Weapons	*3-2-607
Sight Parallelogram Error	3-2-701
Optical Target Acquisition and Tracking Systems	3-2-705
Radio Frequency Radiation Hazards to Electroexplosive Devices	3-2-615
Radio Frequency Radiation Hazards to Personnel	3-2-616
Electrical Power Measurements for Vehicle Subsystems	*1-2-602

(4) Vulnerability.

Protection Against Kinetic Energy Projectiles	2-2-715
Armored Vehicle Vulnerability to Conventional Weapons	2-2-617
Vulnerability to Nuclear Weapons	2-2-618
Resistance to Severe Shock	2-2-620
Vulnerability of Weapons	3-2-531

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b. Specialized applications of weapons mounted on full-tracked carriers are covered in TOP volume III.

SECTION III
SUPPLEMENTARY INSTRUCTIONS

5. Preparation for Test

a. Review of Test Guidance.

(1) The subtests that are selected by the test agency are governed by requirements in the ROC's, MN's, and TC's and by test directives from supervising agencies. The levels of acceptable performance are usually stipulated. If adequate test guidance does not exist, the test plan writer will make appropriate inquiries to the responsible directorate at TECOM.

(2) Certain important characteristics, especially those involving advanced engineering technology, may not be specifically covered by the requirements document but should nevertheless be included in the test plan and evaluated as a part of development test I, II, or III, per AR 1000-1 (formerly the developmental suitability test, engineering test, and initial production test). These may include such features as infrared emissions, shielding against fallout radiation, decontamination of CBR agents, noise measurements, transient voltages, and electromagnetic interference and emissions.

b. Safe Operations During Testing. All hazardous test operations must be covered by a standing operating procedure (SOP) that will provide compulsory safety measures to be followed (e.g., as required by APG Regulation 385-1 at Aberdeen Proving Ground). Some typical operations that are categorized as hazardous are: amphibious operations, carrying bulk quantities of fuel, longitudinal and side slope operation, opration in inclosed chambers (such as cold rooms), and transportation of explosives. Safety measures for routine operations are prescribed in local safety manuals. Lesser routine hazards are covered by internal operating procedures. If a hazardous test is contemplated, the test director should assure that a suitable SOP covers the testing. If none is available, he must write one and obtain approval for its use.

c. Sequence and Selection of Subtests. The actual sequencing of subtests should consider performing the high-risk, short-duration tests early and the low-risk, long-duration tests later in the test period. When required by a TECOM directive, a formalized assessment of the risk associated with each subtest will be made in accordance with TECR 70-34. The high-risk subtests are those in which a failure may lead to early termination of the vehicle test or justify an early judgment on critical performance parameters.

(1) Such items as stowage compartments, fording kits, cold weather kits, maintenance test packages, heaters, and air conditioners

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can be utilized and at least partially tested while the vehicle is undergoing endurance testing. Certain tests, such as those involving vulnerability and accident safety, should be scheduled late in the test program because of the possibility of damaging the vehicle. The type of test, availability of certain kits and similar equipment, and the nature of the possible effect of the test phase on the vehicle or equipment being tested are also factors. A safety evaluation, for instance, would be conducted immediately after the initial inspections and preliminary operation so that the safety aspects could be disseminated to interested agencies. Performance tests covering (e.g.) standard obstacles, drawbar pull, cooling, engine cold starting, and mobility should be conducted prior to or at intermediate endurance testing mileages. It is desirable to obtain the performance data before the engine, drive train, and chassis are subjected to the rigors of extensive endurance testing but after adequate run-in. Comparisons of the performance of various vehicles over a number of years will thus be more valid since all will have been tested in generally the same physical condition.

(2) To test and evaluate a full-tracked carrier a number of subtests apply. The selection of appropriate tests is dependent upon the type of test (e.g., development test I), the availability of test resources (e.g., number of vehicles, funding, test facilities), the time available for the test, the special characteristics of the vehicle, and the desires of the agency overseeing the test program. The possible test phases that may be included in test plans are listed in paragraph 4; rarely will a vehicle be subjected to all. Certain phases of the development tests may require participation by the Boards just as in the past some engineering test phases were combined with the service test when directed by Headquarters, TECOM.

d. Sample Size. The test sample size is usually determined far in advance of testing by the number of vehicles on a production contract, the funds available for testing, or the nature of the test. When the test director has an input regarding the sample size, he should use available mathematical procedures, taking cognizance of stipulated confidence and reliability levels as explained in paragraph 9. Miles that each vehicle must travel are determined simultaneously with sample size.

e. Familiarization and Training. When a vehicle introduces new esoteric features, it may be necessary to train operators, maintenance personnel, and the test director. Such training will normally be given by TACOM or a contractor, and should be arranged at an early date.

6. Initial Inspection and Servicing.

a. An initial inspection is conducted to assure that the test items are in good condition and that major component serial numbers and other pretest data are recorded prior to initiation of the test program. These include:

(1) Serial numbers and other identification of vehicle and major components.

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(2) Examination of maintenance test package for completeness.

(3) Electrical system output (per TOP/MTP 2-2-502).

(4) Odometer mileage and engine-hour readings.

(5) On-vehicle equipment (OVE) stowage (per TOP/MTP 2-2-802).

b. The vehicle is completely serviced to include draining of fluids and refilling with standard fuel, lubricants, and other fluids. The identification and quantity of all fluids used are recorded, and attention is called to unusual quantities. If drain or fill times are unreasonably long, these should be noted together with information on influencing factors such as size of drain plugs.

7. Vehicle Characteristics. Vehicle characteristics including physical dimensions are recorded in accordance with TOP/MTP 2-2-500. A characteristics photograph is taken, and a tabulation of important features is assembled also in accordance with TOP/MTP 2-2-500.

8. Safety Evaluation. An interim safety release should be generated within 60 days of test initiation. The interim safety release recommendation is given based on subtests performed early in the test. The subtests include, but are not limited to, preliminary operation, braking, steering, center of gravity, gradeability, side slopes, toxic hazards, winches, speed, obstacles, and night operations. The safety evaluation does not end with the issuance of the safety release: it is considered to be a continuing subtest throughout the test program and, when required, will include vehicle collision and accident safety.

9. Endurance, Durability, and Reliability.

a. Endurance is a general term that is concerned with the ability of a vehicle to perform satisfactorily under typical field conditions for long periods of time. Endurance tests of tracked vehicles are covered in TOP/MTP 2-2-507 which prescribes test courses and number of miles unless otherwise prescribed by the directive or guidance document.

b. Durability is a precise term that relates to the mathematical probability that a vehicle will be able to operate under typical field conditions for a specified number of miles before requiring major overhaul. Since durability tests require many samples and considerable funding, they are seldom conducted; the endurance test, which is much more limited, is usually conducted instead. TOP 1-2-502 describes a durability test and provides guidance for selecting sample sizes and distances.

c. Reliability is defined as the probability that an item will perform its intended function for a specified time under specified conditions (i.e., the probability of successfully completing a mission

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without incurring a mission-aborting failure). The reliability requirement is usually expressed as a probability of success for one or more specified operational and environmental cycles or functional sequences (mission). This requirement may be expressed in several ways; e.g., as an acceptable mean-time-between failure (MTBF). In the MTBF context "T" may be expressed as hours, cycles, rounds, miles, etc. Most of the data for the reliability determination are obtained from the endurance test, which simultaneously provides data for the maintenance evaluation.

d. The matter of endurance, durability, and reliability is more fully covered in TOP/MTP 2-1-001.

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APPENDIX
REFERENCES

1. AR 70-38, "Research, Development, Test, and Evaluation of Materiel for Extreme Climatic Conditions."

2. AR 705-25, "Reliability Program for Materiel and Equipment."

3. AR 1000-1, "Basic Policies for Systems Acquisition by the Department of the Army."

4. AMC Supplement 1 to AR 70-10, "Test and Evaluation During Research and Development of Materiel."

5. AMC Supplement 1 to AR 705-50, "Army Materiel Reliability and Maintainability."

6. AMCR 385-12, "Life Cycle Verification of Materiel Safety."

7. AMCP 702-3, "Reliability Handbook."

8. AMCP 706-134, "Maintainability Guide for Design."

9. TECR 70-23, "Equipment Performance Reports."

10. TECR 70-34, "Risk Analysis for Suitability Tests."

11. TECR 385-6, "Verification of Safety of Materiel During Testing."

12. TECR 750-15, "Maintenance Evaluation During Testing."

13. MIL-STD-721B, "Definition of Terms for Reliability Engineering, Human Factors, and Safety."

14. Bekker, M. G., "Introduction to Terrain Vehicle Systems," The University of Michigan Press, Ann Arbor, Michigan.

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